The Law of Graphic Equalisation — Information Gravity Grounding:

Change-Load, Quantum Realisation, and the Reconciliation Programme

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DRAFT — draft-2026-09-06.1 r4 (6 September 2026). This is the *01h* grounding companion to *The Law of Graphic Equalisation — Theory*, *Physics Grounding*, *Born Recovery*, *Total Perspective Vortex Grounding*, and *The Law of Narrative Momentum*. It presents a falsifiable *reconciliation programme*, not a completed solution to quantum gravity. Its positive result is a typed bridge ledger: it says what a physical realization must supply before the project can claim a common quantum–gravitational domain. It derives no metric, Einstein equation, Newton constant, Lorentz factor, numerical force law, dark-matter phenomenology, or successful quantum-gravity theory. Physical gravity is treated as one possible grounding of a more general relational-information principle; self-grounding and cross-modal recurrences remain independent structural witnesses with separate tests, not evidence for physical gravity.

Abstract

Graphic Equalisation supplies a relational vocabulary for possibility, frame-indexed realization, history, and change cost. Its physics companion proposes a common change-budget shape for motion and change-load; its Born note maps the vocabulary to a conditional quantum-reconstruction programme. Neither map supplies a physical state space, gauge-invariant record, conserved source, metric, field equation, exact clock law, or empirical discriminator. This paper makes that boundary operational. It defines a common relational realization situation, separates inherited statements from new physical hypotheses, and sets out the Q1–Q6 quantum and G1–G8 gravitational bridges a concrete model must close. The result is a programme that can fail: a candidate loses its reconciliation claim if it cannot provide a noncircular quantum composition and Born bridge, a conserved dimensioned source, a controlled geometry and proper-time recovery, compatibility with established limits, and a constrained discriminator from general relativity plus standard quantum-field-theory effective field theory. Related thermodynamic, holographic, information-geometric, causal-set, and gravity-mediated-entanglement work supplies important conditional results and targets; none supplies these missing maps for the present substrate. The paper therefore advances a test protocol, not an asserted unification.

1 Scope and terminology translation

The familiar question is whether a relational-information account can reconcile quantum descriptions with gravity. The answer established here is precise: the current corpus supplies a *programme specification*, not a quantum-gravity solution. A reconciliation requires one model with translation maps and compatible predictions in a shared quantum–gravitational domain. An analogy shares a motif; a recovery instantiates a target formalism under declared assumptions; a derivation proves a target from stated premises. These are not interchangeable achievements.

The project starts from a general relational-information principle. Physical gravity is a proposed grounding of that principle, not its source. This asymmetry matters: a physical failure can reject a proposed physical grounding without deciding every nonphysical structural use of the relational vocabulary. Likewise, recurrence in self-model, cognitive, computational, or institutional cases is a structural witness only when its own comparison class and tests have been supplied. It neither proves a shared physical mechanism nor transports a gravitational result across modalities.

Terminology guard. "Information" below is not thermodynamic entropy, von Neumann entropy, a modular Hamiltonian, energy, or stress–energy. A frame-relative record is not automatically a diffeomorphism-invariant local observable. A graph or partial order is not a Lorentzian manifold; a statistical or Fisher metric is not the dynamical spacetime metric; and an entanglement pattern is not metric dynamics. Each identification would require a stated map and an independently checkable domain.

2 Existing project premises and the common reconciliation object

The inherited vocabulary can be collected without promoting it to physical ontology. Let the *relational realization situation* be

$$\mathfrak{R}_t = (W, G_t, \Gamma_t, \mathcal{P}_t, \mathcal{H}_t, \mathcal{C}_t, \mathcal{O}_t), \qquad G_t = (g_t, r_t, \Gamma_t),$$

where W is an individuating frame relation; $\mathcal{P}_t = \mathrm{Poss}(G_t)$ is the admissible-continuation family; $\mathcal{H}_t$ is realized history when it is represented; $\mathcal{C}_t$ is change-cost bookkeeping; and $\mathcal{O}_t$ is the relevant frame-held record/readout relation. The inherited realization relation is

$$\mathrm{Coll}_t^W : \mathcal{P}_t \rightsquigarrow G_{t+1}.$$

This is a typed common schema, not a state space, action, path integral, causal-set dynamics, or observable algebra.

Proposition 2.1 (Frame-indexed realization)**.** *Under the corpus definitions, realization is indexed to the frame holding the relevant interaction and record. No frame-independent global realization time follows from this index.*

Proof. The possibility family, admissibility grammar, record, and represented successor are all defined relative to the antecedent graph as held through W. Erasing W erases the relation that individuates the update. This establishes a typed internal consequence only; it does not establish covariant measurement, no-signalling, complete positivity, gauge invariance, or an observer-independent causal order. □

The object is useful because it prevents category drift. Possibility has a slot in $\mathcal{P}_t$, change-load in $\mathcal{C}_t$, local succession in realization order, and physical geometry in a future map from $\mathfrak{R}_t$. It is insufficient because none of those slots fixes amplitudes, a conserved source, proper time, or a metric. The history and cost slots inherit their limited roles from *The Law of Narrative Momentum*: operative history and alteration burden are not universal physical energy. Their comparison-class, support, unit, and ordering discipline inherits from *Total Perspective Vortex Grounding*. Neither companion supplies a physical source or observable; both prevent that missing bridge from being hidden. The table records the paper's claim ledger before any bridge is named.

Derived / inherited claims	New hypotheses	Open bridge obligations	Empirical predictions / constraints
A typed relational schema can be assembled; realization is frame-indexed; admissible possibility does not determine amplitudes.	H-QA: a physical amplitude/state representation exists for the schema.	Define composition, transformations, normalization, and gauge-invariant instruments.	The resulting probabilities must meet the stated quantum tests; a competing non-Born theory satisfying the premises rejects a claimed Born derivation.
Change-cost is a grounding-dependent relation, not an identified physical source; cycle order is not proper time.	H-CL: change-load maps to a dimensioned conserved effective source. H-CT: one clock map yields both tested dilation limits.	Supply source units, exchange/conservation law, geometry, dynamics, and clock observable.	Weak-field, equivalence, local-Lorentz, redshift, Shapiro-delay, and precision-clock tests constrain or reject the map.
Relational graphs do not determine Lorentzian geometry or dynamics; 01d remains conditional at its linearity/composition seam.	H-GEO: a controlled coarse-graining yields physical geometry and observables.	Recover a continuum/alternative, metric/connection or replacement, QFT/EFT overlap, and black-hole scope.	A candidate must state a finite parameterization and at least one result distinguishable from GR plus standard QFT EFT; a null or contrary result must exclude it.

3 Physics grounding: change-load and existing relativistic limits

The physics companion proposes a fixed change-budget picture: motion and internal change-load are two uses of one throughput. That is an information-level shape intended to organize a reading of kinematic and gravitational clock behavior. It is not a physical source equation. In particular, the existing project has not derived a metric, a connection, G, Einstein equations, a Lorentz transformation group, the exact Lorentz factor, a gravitational-redshift law, a numerical force law, or dark-matter phenomenology.

Proposition 3.1 (No source identification follows)**.** *The inherited alteration-cost and change-load vocabulary does not define an energy density, momentum flux, stress tensor, modular energy, entropy, or source for curvature.*

Proof. The cost currency, payer, and comparison class vary by grounding. A physical source requires units, an operational measurement, a locality or nonlocality domain, a conservation or exchange law, and a coupling rule. None is fixed by $\mathcal{C}_t$ or by the budget image. Thus no insertion into a gravitational field equation is typed by the present premises. □

Proposition 3.2 (No clock law follows)**.** *Local realization order or closure-cycle counting does not entail either the special-relativistic Lorentz factor or gravitational redshift.*

Proof. An order admits many inequivalent clock functions. The present corpus supplies no metric, worldline functional, transformation group, or physical clock observable that identifies an operational counter with measured proper time. In particular, a closure-count is only the default simple clock hypothesis: the corresponding execution may be locally closure-driven, externally clocked, or scheduled by a superframe without an internally available discriminator. □

Hypothesis 3.3 (H-CL: load–source bridge)**.** *A candidate realization supplies a map from $\mathcal{C}_t$ to a dimensioned effective source $T^{(\mathrm{eff})}_{ab}$, or to an explicitly different named source, with operational definition, coupling, and conservation or accounted exchange law.*

Hypothesis 3.4 (H-CT: unified-clock bridge)**.** *A candidate realization defines a physical clock observable and proves, in its claimed regime, both the tested kinematic and gravitational clock relations, or states an experimentally distinct alternative with its measurement protocol.*

These hypotheses are hard constraints, not labels for a desired answer. H-CL fails if no source can meet the required conservation compatibility, or if a nonuniversal coupling violates established equivalence or local-Lorentz bounds. H-CT fails if its physical clocks disagree with established dilation, redshift, Shapiro-delay, post-Newtonian, or precision-comparison results in the regime it claims. The empirical review literature documents how tightly metric-gravity and universality departures are constrained (Will, 2014).

4 Quantum possibility, realization, and what 01d supplies

The Born note supplies a correspondence programme between project vocabulary and operational/reconstruction conditions. It identifies a classical probability backbone and marks complex structure, composition, and linearity assumptions or residuals. This paper carries that boundary forward rather than treating a quantum vocabulary as a completed noncircular derivation.

Proposition 4.1 (Possibility is not amplitude)**.** *From G_t, Γ_t, and $\mathcal{P}_t = \mathrm{Poss}(G_t)$ alone, no complex amplitude representation follows.*

Proof. Membership in an admissible set supplies neither a scalar field nor addition, phase, inner product, normalization, composition, or dynamics. Inequivalent probability theories can be defined over the same alternative set. □

Proposition 4.2 (Conditional Born boundary)**.** *If a realization supplies quantum state structure, a suitable noncontextual probability measure, and every dimensional and composition assumption of an applicable reconstruction result, then Born weights can follow within that result's stated domain. The present corpus does not establish those premises in full.*

Hypothesis 4.3 (H-QA: quantum representation bridge)**.** *A candidate realization provides an operational state/effect space, an amplitude or explicitly alternative representation, sequential and composite composition, and normalized frame-transformation rules without importing the conclusion it claims to derive.*

Quantum information tools cannot close this gap merely by being invoked. Holographic codes, information geometry, and gravity-mediated-entanglement protocols ordinarily use quantum state spaces and Born statistics as input. A claimed Born bridge fails if it assumes Hilbert linearity, the trace rule, or standard composition while announcing those structures as outputs. A physical record bridge also fails if the record changes under gauge redundancy: gravity makes local observables and subsystem factorization nontrivial (Donnelly and Giddings, 2016).

5 The joint bridge map and exact unclosed obligations

A reconciliation candidate must publish a concrete realization $\mathfrak{S}$ of $\mathfrak{R}$, not a prose analogy. The following maps are the minimum interfaces:

$$Q_{W,t} : \mathcal{P}_t \longrightarrow \mathcal{S}(\mathcal{A}_W), \qquad A_{W,t} : \mathcal{P}_t \longrightarrow \mathcal{H}_W,$$
$$\Phi_\epsilon : \mathfrak{R} \longrightarrow (M, g_{ab}, \nabla, \mathcal{O}_{\mathrm{phys}}), \quad \Theta_W : \mathfrak{R} \longrightarrow \tau_W.$$

Here $\mathcal{S}(\mathcal{A}_W)$, $\mathcal{H}_W$, (M, g_{ab}, ∇), and τ_W are targets to be defined by the candidate, not objects supplied by the corpus. The amplitude codomain may be replaced by another operational theory, but then the alternative must state its composition, probability, and test domain.

Quantum obligations. Q1 defines operational state/effect spaces and normalized affine probabilities. Q2 defines sequential and composite composition, including any gauge-constrained replacement for naive factorization. Q3 derives or admits a complex Hilbert/C*-algebraic structure with transformations and composition. Q4 derives Born weights from Q1–Q3 plus all necessary assumptions, or openly admits them as an input. Q5 makes records and realization events gauge-invariant relational observables or instruments. Q6 specifies unitary or nonunitary domain, normalization, and strong-gravity or black-hole scope.

Gravitational obligations. G1 supplies a controlled coarse-graining to a Lorentzian continuum or stated alternative with causal observables. G2 identifies metric/connection data or an alternative and their transformation law. G3 closes H-CL. G4 supplies a field equation or tested alternative with error and domain. G5 derives or parameterizes universality and local Lorentz behavior. G6 closes H-CT. G7 derives entropy/area and black-hole behavior or limits the domain explicitly. G8 identifies how gauge-observable and relevant no-go assumptions are satisfied or evaded.

Conditional compatibility target T0. Let $\mathfrak{S}$ be a concrete realization. If it closes Q1–Q6 and G1–G8, preserves the claimed operational predictions in their shared domain, and publishes a discriminator from GR plus standard QFT EFT, it is a candidate reconciliation in that domain. T0 is a target specification rather than a theorem of this paper: no concrete $\mathfrak{S}$ has been supplied, so none of its antecedents has been proved.

6 Relation to prior art

Thermodynamic and entropic routes. Jacobson derives the Einstein equation as an equation of state under a local horizon Clausius relation, area-entropy, and stress-energy-flux assumptions (Jacobson, 1995). His entanglement-equilibrium route adds fixed-volume causal-diamond and quantum input (Jacobson, 2016). These are conditional gravitational results, not a derivation that $\mathcal{C}_t$ is entropy, energy, or curvature. Verlinde's entropic-force proposal is also a proposal with additional screen, temperature, and equipartition assumptions (Verlinde, 2011); implementations motivated by later emergent-gravity work faced empirical mismatches in isolated dwarf systems (Lelli et al., 2017). This paper claims neither their force law nor their phenomenology.

Holography and quantum error correction. AdS/CFT is a conjectured gauge/gravity duality with extensive evidence in particular settings (Maldacena, 1998). In static settings, Ryu–Takayanagi relates boundary entropy to a minimal bulk surface under its own conditions (Ryu and Takayanagi, 2006); the covariant Hubeny–Rangamani–Takayanagi extension uses extremal bulk surfaces (Hubeny, Rangamani and Takayanagi, 2007). Entanglement-wedge reconstruction has quantum-error-correcting structure in a code subspace (Almheiri, Dong and Harlow, 2015). These furnish exact examples of encoding and reconstruction, not evidence that an arbitrary relational graph produces generic spacetime, quantum amplitudes, or metric dynamics.

Information geometry and causal sets. Information geometry gives metrics on statistical models (Amari, 2016); those metrics are not thereby Lorentzian spacetime metrics. Causal-set theory gives a concrete locally finite partial-order programme for spacetime (Bombelli et al.,

1987; Surya, 2019). It is a useful target because causal order and volume provide more physical structure than the present graph vocabulary, but it does not itself supply this project's Born, budget, or cross-modal bridges.

Quantum-gravity witnesses and constraints. Gravity-mediated entanglement proposals test nonclassical mediation only conditional on their specified locality and mediator assumptions (Bose et al., 2017; Belenchia et al., 2018). Even a successful witness would not select this substrate, detect a graviton, or validate a unique quantum-gravity theory. Any emergent massless-spin-two claim must also identify its relation to the assumptions of Weinberg–Witten (Weinberg and Witten, 1980); "emergence" alone is not an evasion. At low energies, quantum general relativity is usefully organized as an effective field theory (Donoghue, 1994), which a candidate must recover or delimit.

7 Empirical discriminators, recovery conditions, and falsifiers

No numerical prediction is claimed because no realization fixes a source normalization, continuum scale, parameter, or observable. A candidate earns a calculation only after it gives a quantity with units, parameterization, uncertainty/control analysis, and comparison to GR plus standard QFT EFT in the same domain. Dimensional rhetoric is not a prediction.

Recovery conditions. The classical recovery condition is a specified low-velocity, weak-field, decohered limit with source law and force, potential, or geodesic equation. The special-relativistic condition is covariant observables and the exact tested clock relation. The general-relativistic condition is a metric/effective dynamics with weak-field, redshift, light, and post-Newtonian agreement. The quantum condition is a state space, composition, dynamics, instruments, and Born statistics. The semiclassical condition is a renormalized observable and controlled backreaction domain. None is presently obtained by the corpus; each is a required future calculation.

Failure protocol. The programme direction fails as a claimed reconciliation for a candidate if any of the following occurs:

1. no noncircular quantum-composition and Born bridge can be constructed;
2. no dimensioned, conserved source or accounted exchange law maps change-load to physical dynamics;
3. no continuum/alternative map supplies gauge-invariant causal and clock observables compatible with local-Lorentz and equivalence constraints;
4. the candidate conflicts with established clock, weak-field, or EFT predictions within its stated regime;
5. it meets a no-go theorem's assumptions while asserting the prohibited object; or
6. all proposed realizations remain observationally identical to GR plus standard QFT EFT and provide no further finite, rejectable parameter claim.

The final item is not a demand for novelty theatre. It distinguishes a compatible reinterpretation from a theory with additional empirical content. A valid proposal must name the observable, parameter region, comparator, protocol, and null or contrary outcome that excludes its mechanism.

8 Conclusion

The project currently supports one hard result: a reconciliation claim has a checkable type. A physical candidate must translate frame-indexed realization into gauge-invariant instruments, admissible possibility into a quantum representation without circular Born input, change-load into a conserved physical source, and relational structure into controlled causal geometry and proper-time observables. It must then recover the domains where GR and quantum field theory already work and risk a constrained disagreement.

Until those bridges are published for a concrete realization, this is not a quantum-gravity solution. It is a reconciliation programme with named obligations and failure conditions. Physical gravity can ground one realization of the wider relational-information principle; it cannot be used as that principle's source, and nonphysical structural witnesses must continue to earn their status under their own tests.

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